P1 - Topic & Objectives: Appointment Booking System

Mission Statement

This appointment booking system would exist to provide businesses that typically offer services through one-on-one timed sessions with their clients. Some of these businesses include hair & nail salons, medspas, pet grooming, and more. This database design would allow customers to book appointments on a digital platform at any time. This would also assist customer-facing staff, allowing them to focus more on helping clients on site. As this is a client-facing design, its focus lies on making a seamless booking experience for the customer. Although additional information could be used from these bookings to provide business insights such as sales reports, trends, reviews, etc.; however, this would be outside the scope of this design. In other words, the design will be scalable enough to provide such information in the future.

Mission Objectives

- Provide customers a way to digitally book their appointments, whenever is most convenient for them
- Provide customers information about the business, such as physical location, operating hours, contact information
- Customers will have an account tied to their phone number and email address
- Provide available slots for these appointments, based on staff availability and the kind of services they are able to provide
- Each service will have an estimated time slot and price
- Customers would be able to change their appointment as well as cancel
- Each appointment will have a cancellation policy and fee set by the business
- Cancellation fees would be charged on a valid credit card
- Credit cards may be optionally stored for future bookings
- Send a confirmation request to customers as well as optional reminders for their appointments

Main Entities

1. Customer: the person booking
2. Business: the company offering appointments
3. Location/Branch: if the business has more than one site
4. Staff/Provider: the person offering the service
5. Service: what's being booked
6. ServiceCategory: groups related services
7. Appointment: the booking itself, which would link customer, staff, service,estimated charge and time slot
8. StaffAvailability: staff working hours, blocked-off time

9. TimeSlot: discrete bookable windows, often derived from availability
10. Deposit: charge tied to an appointment; Modeled narrowly as Deposit to match current scope; could generalize to a Payment entity with a type attribute if the platform later supports full checkout
11. Cancellation: Records that a specific appointment was cancelled (when, by whom, and what happened to the deposit as a result of applying the relevant policy at that moment in time)
12. CancellationPolicy: Defines the rules a business sets for how cancellations are handled, how much notice is required and what fee or refund applies if that notice isn't given. Set once per business, and reused across many appointments
13. Notification: confirmation, reminder, cancellation alerts
14. StaffService: which staff can perform which services